

to other charging solutions. The I²C interface in bq51201 enables designers to implement the pad detection feature, an easy way to align receiver with the charger and the foreign object detection to allow faster and efficient charging of all the Qi-compliant gadgets.

Lesser components

Texas Instruments also came up with bq500412 that allows designers to develop charging stations with half the number of components used in other charging solutions. This transmitter, featured on foreign object detection though created for 12V charging pad, can also be used with a boost converter to create a 5V USB charging pad with three coils to limit power loss.

PowerbyProxi brings a fully integrated receiver circuit that fits easily into a smartphone to charge its battery wirelessly. You can place at least three devices on its pad for simultaneous charging. It charges a device at the speed of a wired charger without overheating its components.

Comforting power

Charging has moved on from what has hitherto been a utility to one that comes with an aura of comfort. Though the modules integrated in furniture are based on Qi technology they can also charge the devices that are not based on Qi technology. To accomplish this they use an adaptor with the module. C+P Furniture Systems is working to bring the whole new Cegano smart table and Prefino locker system with the wireless charging facility for smartphones.

Leggett & Platt have come up with Helios, an inductive charging system that is available in puck shapes or low-profile designs. It is an intelligent system that communicates with the device to be charged. That is, the primary coil communicates with the secondary to identify its power requirements and charges it accordingly. Once Helios identifies that the device is fully charged, it cuts off charging and turns on the LED indicating full charge. Although the device is charged wire-

TABLE I Technical Analysis of Standards			
	EPC	PMA	A4WP
Freq	100-205 kHz	250-400 kHz	6.78 MHz
Range of coupling	K-0.4 to 0.7	K-0.6-0.8	K-0.1 to 0.5
Q of Coils	6-10	6-10	40-100
Communication	In Band (detailed)	In Band (simple)	Out of Band-BLE
Alignment	Magnet on TX (retired)/ Multi Coil	Magnet on TX and RX	No alignment needed
Coil size/cost	Med/Med	Small/high	Small/small
Thickness	0.3-1.0 mm	0.8-1.3 mm	0.2-0.5 mm (Est)
Semiconductors RX	<20V	<20V	40V-60V
Architecture of RX	Rect + Reg + internal COMM	Rect + Reg + internal COMM	Rect + Buck _ BLE
Heating of metal	Med (FOD implemented)	Med (will need FOD in future)	Low
Form factor flexibility	Med	Low	Med-High
Efficiency	70%+	70%+	~50%

Courtesy: Texas Instruments

TABLE II Market Analysis of Standards			
	WPC	PMA	A4WP
Member companies	154 + members	86 + members	55 + members
Active consortium	Yes	Yes, formal technical	Yes
Specification published	Yes WPC 1.1, public	Yes PMA 1.1, members only	Yes V1.0, members only
Approved Tx types	20	1	N/A (3 in development)
Test houses established	9	None	None
Certified products	231	None	None
IC Solution available	Yes (~5 Suppliers)	None public	None public
Magnetics Solution available/understood	Yes	Yes, limited	No
Regulatory approvals	Yes	Yes	No
Infrastructure play	Japan, Europe	US, Europe	None
Active carriers	Verizon and Docomo	ATT	Tmobile
Current shipment vol	15M+	<1M	N/A
Comments	Well established, products in market, wide design flexibility	Dominant US player, Strong ATT support, early stage	Strong push by SS and Ocom, Active development ongoing

Courtesy: Texas Instruments

lessly, Helios has to be plugged into a socket to power the primary coil.

"Wireless charging furniture tables, chairs and desks—is not very far. In fact, many of us (including Knewron) are already in design phase of these things. However, these are not out-of-the box solutions either; just being manifestation and augmentation of

existing technology with surroundings. Either way, it will soon be ready to join the interior decoration and furniture market," explains T. Anand, Knewron.

New standards

Alliance for Wireless Power (A4WP) is a wireless charging standard that came up recently. Although the products